A Preregistered Falsification Program for a Coherence-Inheritance Control Framework: Simulation Testbeds and Two Hardware Tests

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Abstract. We report the complete and closed Phase 1 record of a falsification program applied to a framework originally proposed as "Unified Inheritance Physics" (UIP) — a candidate formalization of coherence, drift, inheritance, and adaptive correction as control-theoretic and information-theoretic objects. Rather than advocating for the framework, we subjected its claims to nine preregistered tests with explicit kill-conditions, publishing every outcome the same day. Final ledger: **four claims confirmed, five falsified**. Confirmed: a single-qubit recirculation fixed point with steady-state drift predicted analytically to within 2% in simulation (Appendix A); a restricted-access regime in which inheritance-guided control is advantageous, together with a quantitative crossover law $s^*(\gamma)$ confirmed as preregistered (Appendices C–E). Falsified: inheritance-guided two-qubit control against a static baseline in two preregistered versions (Appendix B); a control-sufficiency claim against a near-optimal policy gauntlet, gap 12.19% against a 5% threshold (Appendix F); the universality claim itself, leading to formal retirement of the word "Unified" (Appendix G); and the fixed-point law on real hardware — twice. On IBM Quantum's ibm_kingston processor, the law with published-calibration inputs was rejected at 383.7% deviation against a preregistered ±10% window (Appendix H); after correcting the diagnosed T2/T2* input error with echo-protected idles, the law was rejected again at 81.9% deviation (Appendix I), which per preregistration falsifies the law itself, not its inputs. The qualitative fixed-point structure — fast convergence to a flat, stable coherence plateau — was observed in both hardware runs; the quantitative level requires loss channels (gate error, readout error, T1) absent from the two-parameter model. Phase 1 hardware testing is closed under the program's no-third-attempt rule. Nothing in this record is claimed as new physics; all results reduce to known quantum information theory and feedback control. The contribution is methodological: a complete, public, same-day-published ledger in which negative results carry equal weight — including two hardware rejections of the author's own predictions — produced by an independent researcher using open infrastructure.

1. Introduction and honest framing

The framework under test began as a broad proposal: that coherence C, drift D, resonance R, inheritance, and adaptive correction could be formalized as measurable objects obeying a common operating equation across domains. Phase 1 was defined from the outset not as an advocacy exercise but as a falsification program: each claim was preregistered with an explicit numerical kill-condition before any test was run, and every outcome — pass or fail — was published the same day in a public repository and archived with a DOI.

Two framing commitments govern the entire record. First, **no claim of new physics is made**: every confirmed result is consistent with, and reducible to, standard quantum information theory and classical feedback control. Second, the framework's original universality claim did not survive testing (Section 4.4), and the word "Unified" was formally retired from the project's name as a consequence. What remains after closure is a candidate control-design heuristic with sharply mapped boundary conditions in simulation, a fixed-point law demonstrated to be a simulation result rather than a hardware law, and a methodological demonstration.

2. Framework definitions

The Phase 1 objects, instantiated numerically on qubit systems:

- **Coherence** $C(\rho)$: the ℓ1-norm coherence (magnitude of off-diagonal elements) of the system state in the computational basis; on hardware, the proxy $C = |\langle X\rangle|$ for a $|+\rangle$ target.
- **Drift** D: distance from the target state; for the single-qubit testbeds, $D = (1 - C)/2$.
- **Resonance** R: coupling/transfer rate between subsystems (e.g., partial-SWAP strength).
- **Inheritance** $I(A \to B)$: coherence deliverable from an upstream to a downstream system; tested definitions included $I = R\cdot C(A)$ and $I_2 = R\cdot C(A)\cdot\tau\_eff$.
- **Operating equation:**

$$dC/dt = G(R, U) - L(D, W)$$

with G a gain functional of resonance and applied correction U, L a loss functional of drift and waste W. The central discrete-time result used throughout is the recirculation fixed point: under per-cycle decay $e^{-\gamma}$ and partial re-preparation with gain f,

$$C^* = f / (1 - (1-f)\cdot e^{-\gamma}), \quad D^* = (1 - C^*)/2$$

3. Method: preregistration with kill-conditions

Every test followed the same protocol: (i) hypothesis and numerical kill-condition committed to the public repository before execution; (ii) test executed exactly as specified; (iii) outcome published same day regardless of result; (iv) post-hoc analyses labeled as such and assigned zero evidential weight; (v) failed claims either retired or sharpened into new preregisterable hypotheses — never silently retried. A strict **no-third-attempt rule** was applied to every falsified claim within Phase 1, to prevent curve-fitting; this rule ultimately determined the program's closure (Section 4.5).

4. Results

4.1 Appendix A — Single-qubit recirculation fixed point (simulation): PASS

Testbed: one qubit holding $|+\rangle$ under dephasing and amplitude damping. A key negative result preceded the pass: **unitary correction alone failed** the drift-recovery and contraction tests (mean contraction factor $k \approx 1.05$) — coherence lost to the environment cannot be restored by rotation. Only *recirculation* (measurement-based partial re-preparation toward the target) succeeded, giving physical content to the framework's slogan that "waste is inheritance in the wrong form." With recirculation, the adaptive strategy passed drift-recovery, beat the generic echo baseline, respected the non-harm envelope, and — the quantitative core — the analytic steady state $D^* = 0.1032$ matched simulation at 0.1051 (<2%). Contraction ($k<1$) passed at moderate gain ($f = 0.5$) from a maximally mixed start.

4.2 Appendix B — Two-qubit inheritance-guided control: FAIL (two preregistered versions)

Claim: the inheritance quantity $I(A \to B)$ identifies where to spend a limited correction budget. Version 1 ($I = R \cdot C(A)$) lost to the static always-protect-the-memory baseline by 11.5% in mean coherence, including under time-varying noise and noisy estimates. Version 2 preregistered a refined quantity $I_2 = R \cdot C(A) \cdot \tau\_eff$ with two predictions: P1 (weak-coupling equivalence to always-B) passed; P2 (existence of a coupling regime where v2 beats all baselines) failed across the full sweep. The falsified lesson: when the target is directly accessible, protecting it directly dominates; upstream coherence value was systematically overweighted. A v3 hypothesis (advantage only when the target is *not* directly accessible) was recorded but deliberately not run in the same cycle.

4.3 Appendices C–E — Restricted access and the crossover law: PASS

The sharpened hypothesis from Appendix B was preregistered and tested: in architectures where the memory qubit cannot be corrected directly, inheritance-guided upstream control becomes advantageous. This passed as preregistered (Appendix C). Appendices D–E extended it to a quantitative, preregistered **crossover law $s^*(\gamma)$** — the coupling threshold at which upstream control overtakes the accessible-target strategy as a function of noise rate — whose predicted functional form was confirmed in simulation. This is the framework's strongest surviving positive content: not a universal principle, but a mapped regime with a predictive boundary.

4.4 Appendices F–G — Control sufficiency and universality: FAIL

Appendix F (optimal-policy gauntlet): the inheritance-guided policy was tested against a near-optimal benchmark policy with a preregistered tolerance of 5%; the measured gap was 12.19% — the heuristic is not close to optimal even in its favorable regime. Appendix G tested the universality claim directly and falsified it; as a consequence the word "Unified" was formally retired from the framework's name. These two failures bound the framework honestly: it is a domain-specific control heuristic with known suboptimality, not a general law.

4.5 Appendices H–I — Hardware tests on IBM Quantum: FAIL ×2 — the law is falsified on hardware

All prior appendices were simulations executed by the framework's authors. Appendices H and I are the arbiters outside our control: two preregistered runs on IBM Quantum's ibm_kingston (Heron r2, 156 qubits), qubit 140, calibration T2 = 601 μs, gain $f = 0.5$, 40 μs idle per cycle, 8 cycles, 4096 shots.

Run	Protocol	Predicted D*	Measured D*	Deviation	Verdict (±10%)
H1 (Appendix H) — job `d99brisqp3as739tudkg`	Bare idle delays; decay rate from published calibration T2	0.0302	0.1463	383.7%	**FAIL**
H2a (Appendix I) — job `d99h2nd2su3c739keu80`		0.0302	0.0550	81.9%	**FAIL**

	X echo at midpoint of every idle, so echo-T2 governs (per prereg `prereg_h2a_echo_fixed_point.md`)				

H1 diagnosis. The single-cycle collapse (C: 0.994 → 0.678) implied an effective per-cycle decay of ≈0.68 versus the predicted 0.936. Published T2 is measured with echo sequences; bare idles decay at the free-induction time T2* — typically 5–10× shorter. H1 therefore falsified the *input model*, and the sharpened hypothesis H2a was preregistered with an explicit escalation clause: failure of H2a falsifies the law itself, not merely its inputs, with no post-hoc parameter adjustment counting as a result.

H2a outcome. Echo protection worked as diagnosed — deviation fell roughly five-fold (383.7% → 81.9%) and steady-state coherence rose from ≈0.71 to ≈0.89, vindicating the T2/T2* analysis. But the measured D^* = 0.0550 still missed the preregistered window [0.0272, 0.0333] decisively. **Per the preregistration, the idealized fixed-point law is falsified on hardware.** The recorded diagnosis (zero evidential weight): the two-parameter model omits real loss channels — single-qubit gate error accumulated over the added H and X pulses, readout error, imperfect refocusing by a single midpoint echo relative to multi-pulse calibration sequences, and amplitude damping (T1), which echoes do not remove.

What survived both runs: the qualitative fixed-point structure. In H1 and H2a alike, coherence converged rapidly to a flat, stable steady state exactly as the recirculation model predicts — but at a level set by the device's true aggregate loss, not by calibration T2 alone. The law's final Phase 1 status: **confirmed quantitatively in simulation (<2%); falsified quantitatively on hardware at ±10%, twice, at two levels of input honesty; qualitative plateau prediction confirmed on hardware.** An extended model with explicit gate-error, readout-error, and T1 terms is a Phase 2 hypothesis, deliberately not attempted. Under the no-third-attempt rule, Phase 1 hardware testing closed with Appendix I.

5. The complete ledger (Phase 1 closed)

Appendix	Claim tested	Kill-condition	Outcome
A	Recirculation fixed point predicts steady-state drift (simulation)	Six falsification tests incl. quantitative D^* match	**PASS (<2%)**
B (v1, v2)	Inheritance-guided budget allocation beats static baselines	Must beat all baselines	**FAIL (×2)**
C	Advantage exists under restricted target access	Preregistered superiority in restricted regime	**PASS**
D–E	Quantitative crossover law $s^*(\gamma)$	Predicted functional form within tolerance	**PASS**
F	Near-optimality of the heuristic	Gap ≤ 5% vs near-optimal policy	**FAIL (12.19%)**
G	Universality ("Unified")	Independent cross-domain transfer	**FAIL — "Unified" retired**
H	Fixed-point law with calibration inputs on hardware	D^* within ±10% on ibm_kingston	**FAIL (383.7%)**

I	Fixed-point law, echo-protected idles, on hardware	D* within ±10%; failure falsifies the law itself	**FAIL (81.9%)**

Final: 4 confirmed / 5 falsified — both hardware verdicts delivered by a device outside the author's control, both published unedited the same day.

6. Discussion

What survived. (i) The recirculation fixed point as a quantitative simulation result and a qualitative hardware structure; (ii) the restricted-access regime and the crossover law s*(γ) — a genuine, preregistered, confirmed quantitative prediction, albeit within known physics; (iii) the physical necessity of recirculation over unitary correction, discovered through an initial failure.

What died, and what the deaths taught. Inheritance-guided control is not generally advantageous (B), not near-optimal where it is advantageous (F), and not universal (G). Hardware killed the fixed-point law twice: H falsified the calibration-input model and taught the echo-T2/T2* distinction between published calibration and bare-idle reality; I falsified the law itself, teaching that a two-parameter idealization cannot survive contact with gate error, readout error, and T1 on a real device. Each failure produced a sharper boundary or a sharper hypothesis; none was softened or hidden.

Methodological claim. The substantive contribution of Phase 1 is not any single result but the demonstrated protocol: an independent researcher, using only open infrastructure (a free quantum-hardware tier, a public repository, a DOI archive), can run a complete preregistered falsification program — including same-day publication of two hardware rejections of their own predictions — in days rather than the months-to-years such a loop traditionally requires. The failures are what make the passes believable. Closing the program on a hardware falsification, at its preregistered threshold, without a third attempt, is the record's strongest credibility asset.

7. Open items and invitation (Phase 2 candidates — not run in Phase 1)

- **Extended hardware model:** the fixed-point law augmented with explicit gate-error, readout-error, and T1 loss terms, preregistered fresh, on hardware.
- **B-v3** (inheritance advantage only under restricted access, on hardware): recorded, unrun.
- **Independent replication:** all code, preregistration commits, job IDs, and raw cycle data are public. Anyone with an IBM Quantum open-plan account can rerun both hardware tests. Refutations are solicited as explicitly as confirmations, via repository issues.

8. Data and code availability

Repository: `github.com/derekhone/uip-phase1-testbeds` (code, preregistrations, Appendices A–I). Permanent archive: DOI `10.5281/zenodo.21246246` (concept DOI, resolves to latest version). Hardware: IBM Quantum open plan, ibm_kingston, jobs `d99brisqp3as739tudkg` (H1) and `d99h2nd2su3c739keu80` (H2a). License: CC-BY.

"A false balance is an abomination to the LORD, but a just weight is His delight." — Proverbs 11:1

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